

The Need to Address Security Concerns in the Cloud Environment



Security of data is of paramount concern to those in government and industry. This article takes readers through the typical security concerns in the traditional environment, before going on to the cloud environment.

Before we discuss security concerns in the cloud, let us take a small detour to look at the security situation in traditional environments. In today's competitive market, a typical IT environment has fewer capabilities to handle dynamic requirements due to budget constraints and a large user base. To survive in the current state and to plan for long-term benefits, organisations need to reduce costs, increase storage capacity, as well as automate the monitoring of IT environments and the management of resources for disaster recovery and availability, so as to achieve a flexible IT environment.

The following are the challenges in the traditional environment:

- Security systems utilise excessive amounts of resources
- Vulnerability scanning or monitoring tools may not detect vulnerability threats due to the fast growing volume of data
- Accurate and timely information about threats is not widely shared; information about the vulnerabilities remains with a smaller group and not all the stakeholders
- Audit systems are not built to operate on the dynamic nature of data centres
- Frequent failovers occur due to limitations on scalability
- Aggressive cost-cutting plans are degrading security programs at a time when threats are escalating
- Security access privileges and roles are not as per standards

Complexities of the underlying infrastructure in on-premise environments have attracted organisations to the

cloud environment to achieve agility, high availability and reliability. Thus, an organisation's security and compliance requirements also need to be aligned and automated in an agile cloud environment to realise the desired benefits of moving to the cloud.

Security in the cloud

Cloud computing provides the next generation of Internet-based, scalable and distributed computing systems in which resources are offered 'as a service'. Today's IT organisations are under increasing pressure to securely deploy applications in public, private or hybrid clouds. Security in the cloud environment is a major concern. Despite the likely gains achieved from cloud computing, the security model is still not entirely fool-proof and hence it impacts cloud adoption. Traditional security controls and policies are not the best fit for the virtual world. Multi-tenancy, virtualisation and geographical differences have made the boundaries of the security model more dynamic than before. Cloud security is no longer the 'elephant in the room' since various organisations and standards bodies have raised their concerns about it. It is the general public as well as the providers and consumers of cloud services who seem to continue in a state of ignorant bliss. Cloud consumers and service providers must realise that a stitch in time will save nine.

Infrastructure abstraction and lack of visibility in cloud resources can create a number of security and compliance issues. In addition to the usual challenges of developing